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Smartphone embodiment: the effect of smartphone use on body representation

Yue Lin[1,2] · Qinxue Liu[1,2] · Di Qi[1,2] · Juyuan Zhang[1,2] · Zien Ding[1,2]

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Abstract
Tool use promotes tool embodiment and extends human body representation. Based on previous studies about tool embodiment, we aimed to explore whether smartphones could be integrated into body schema and extend body representation. We adapted the stimulus–response compatibility paradigm (Experiment 1a, $n=39$; Experiment 1b, $n=44$), the hand mental rotation paradigm (Experiment 2, $n=39$; Experiment 3, $n=40$), and the forearm bisection paradigm (Experiment 4, $n=40$). The results revealed that participants showed a processing advantage for the picture of a smartphone held, indicating that smartphones could be integrated into the body schema. Moreover, the results showed that the perceived forearm length increased after imagining using a smartphone, indicating the extended body representation. Besides, we found no significant difference between high-frequency smartphone users and low-frequency smartphone users across the four experiments. These results underscored the unique role of smartphones in modifying the body schema and extending body representation, which was more potent than traditional tools and non-smartphones. Our findings provide empirical evidence for extended-self theory applied to smartphones and provide a novel insight into the impact of smartphone use on humans from the perspective of tool embodiment.

Keywords Smartphone use · Body representation · Body schema · Tool embodiment · Extended self

Introduction

Throughout evolutionary history, humans have long embraced tools to free themselves from the limits of the human condition and extend the capabilities of the body self (Ambrose, 2001; Martel et al., 2021). Body representation (also known as body schema) is formed through the ongoing, mainly unconscious integration of proprioceptive signals and multisensory information (Maravita & Iriki, 2004). In the interaction with the tools, the brain constantly encodes, stores, and updates body schema (Maravita & Iriki, 2004; Martel et al., 2016, 2021). During this process, tools could be incorporated into our body schema (Giummarra et al., 2008; Jovanov et al., 2015; Osiurak & Federico, 2021; Weser & Proffitt, 2021). The tool incorporation has been referred to as tool embodiment, which allows us to perceive the tool as a part of our body, thus enhancing the exquisite human mastery of tools (Martel et al., 2016). The phenomenon and mechanism of tool embodiment have been widely discussed in traditional tools such as rakes and tongs (Jovanov et al., 2015; Martel et al., 2016, 2021; Sposito et al., 2012). For example, the use of a rake can lead to the integration of the tools into the body self (Bonifazi et al., 2007; Farnè & Làdavas, 2000; Ronga et al., 2021) and increase the perceived forearm length (Cardinali et al., 2012; Martel et al., 2021; Sposito et al., 2012). Moreover, training using a cane induced an extension of the representation of the hand, forearm, and even the calf, which was not directly contacted with the tool (Sun & Tang, 2019). Overall, tool use promotes tool embodiment and extends our body representation.

As the advanced tool of modern technology, could smartphones also show embodied effect on body representation (i.e., smartphone embodiment)? Indeed, the smartphone is one of the most frequently used handheld tools for humans. According to national surveys, 96% of 18–29-year-olds in America own smartphones (Pew research center, 2021),

✉ Qinxue Liu
qinxueliu@ccnu.edu.cn

1 School of Psychology, Central China Normal University, Wuhan 430079, China

2 Key Laboratory of Adolescent Cyberpsychology and Behavior (Central China Normal University), Ministry of Education, Wuhan 430079, China

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and 99.6% of Chinese Internet users use smartphones to access the Internet (China Internet Network Information Center, 2021). Similar to traditional tools, the smartphone has recently been found to extend the capabilities of the self (Clayton et al., 2015; Hoffner et al., 2016; Park & Kaye, 2019). However, smartphones and traditional tools differ largely in the extensions related to physical properties (e.g., length range). The reachable distance of traditional tools is specifically visible, while that of smartphones is more abstract. The connection between traditional tools and the body is also direct, while the connection between smartphones and the body seems to be indirect and virtual. Although the visible extension of the body from the smartphone is not apparent, the information and connection they provide are actually far more expansive than traditional tools. Given the obvious difference between smartphones and traditional tools in the experience of interacting with the body, which is the key to body embodiment (Maravita & Iriki, 2004; Martel et al., 2016), it remains unclear whether the embodied effect of smartphones is comparable to or greater than traditional tools. Exploring these questions will expand the theoretical scope of tool embodiment and provide nuanced insight into understanding the relationship between human and technological tools.

Smartphone use and body representation

According to Extended-self theory (Belk, 1988, 2013), external objects become viewed as part of self when wearable to exercise power or control over them, just as we might control an arm or leg. As a wearable tool, iPhones were found to have the same effect: When iPhone users were unable to check their iPhones, their heart rate and blood pressure increased, and self-reported sense of extended-self decreased (Hoffner et al., 2016). It suggested that the negative psychological and physical consequences, similar to losing a body part, might be associated with iPhone separation. Qualitatively, Park and Kaye (2019) interviewed 60 heavy smartphone users and found the blurring boundary between the "human self" and the smartphone, providing more direct evidence for self-extension via smartphones. Smartphone users asserted that their phones had become an indispensable part of their perceived self and thus influenced their identity.

How could a smartphone be perceived as a part of the body? Embodied cognition underlines that cognition is constrained by the body we possess and stresses the importance of action for cognition (Borghi & Cimatti, 2010). Our perception of the "self" is also based on embodied experience, which is the sense of being localized within one's physical body (Arzy et al., 2006; Suggate et al., 2019). Thus, smartphone users may have embodied experience when using and manipulating smartphones, especially the embodied experience of their hands. Based on embodied cognition, a previous study found that smartphone use could modify body schema by adapting the hand mental rotation task (Liu et al., 2022). In particular, smartphone users responded faster and more accurately to rotate the hand image with a smartphone held in hand than the hand image with a non-smartphone held in hand.

Second, the sense of reward and control provided by smartphones could also promote smartphones to be incorporated into the user's self. The reward system was sensitive to self-related objects (Krigolson et al., 2013), indicating that the perceived reward was essential for incorporating objects into the self. In addition, a sense of control over the object was also an essential factor for incorporating objects into the self (Christoff et al., 2011; Zhou et al., 2015). Indeed, smartphones could provide users a sense of pleasure, satisfaction, and immersion (Hsiao et al., 2016). Smartphones also enable individuals to gain a high sense of control by the aggregation of functions, personalized and customized content, as well as accessibility and convenience (Ahn & Jung, 2016). Therefore, in the interaction with smartphones, we could experience a high sense of reward and control, which will continuously stimulate the brain to encode the self-objects.

Tasks for exploring the updating of body representation

Since smartphones are handheld objects, we adapted the paradigms of hand representation to explore the relationship between smartphone use and body presentation. First, we adapted the stimulus–response compatibility (SRC) paradigm (Ottoboni et al., 2005; Vainio & Mustonen, 2011). SRC refers to the effect of the spatial congruence between the presented stimulus and the desired response on the individual's processing of spatial information. SRC involves the spatial properties of the stimuli and the responses of the right and left hands, thus used to study hand representation (Nishimura & Michimata, 2013; Ottoboni et al., 2005; Vainio, 2011; Vainio & Mustonen, 2011). The hand-related SRC paradigm presents participants with left- or right-handed picture stimuli and requires them to respond to target stimuli with spatial properties (e.g., arrows) using their left- or right-hand. The representation of left- or right-hand was based on activation of the self-body schema, thus leading to faster responses with the hand that was compatible with the identity of the viewed hand when the hand was presented in egocentric perspective (Ottoboni et al., 2005; Vainio & Mustonen, 2011). In contrast, there was a negative SRC effect when hand stimuli were recognized as the hands of others (allocentric perspective) (Ottoboni et al., 2005; Vainio & Mustonen, 2011). Moreover, SRC studies also suggested that identifying grasp-related features of tools with handles could activate representations of hand movements and produce object SRC effect (Pellicano et al., 2010, 2019).

Therefore, the hand or tool image that triggers a larger SRC effect might be more compatible with the individual's body schema. In the present study, we adapted the SRC paradigm to explore the impact of smartphones on hand representation. We proposed that smartphone-related hand images would trigger a larger SRC effect.

Second, we adapted the hand mental rotation paradigm (Ionta et al., 2007; Liu et al., 2022; Ter Horst et al., 2010). In contrast to the implicit priming of hand position perception in SRC, hand mental rotation required a conscious process involving integrating sensory information with motor movements (Bonda et al., 1995; Ionta et al., 2007; Lorey et al., 2009). The hand mental rotation paradigm requires participants to imagine the presented hand image as their own and judge the hand laterality (Ter Horst et al., 2010). Hand rotation was postulated as an embodied process (Parsons, 1994; Suggate et al., 2019). Thus, a short response time was required when the angle and posture of the presented hand were similar to one's own hand (Bonda et al., 1995; Ionta et al., 2007; Lorey et al., 2009). In addition, mental rotation was faster for objects that indicated a higher level of body-object interaction (e.g., hammer, spoon, scissors), suggesting that the mental rotation of hand-related objects was also embodied (Suggate et al., 2019). Therefore, the hand or hand-related object might be embodied into body schema and thus leading to a faster response time in the hand mental rotation paradigm. Indeed, a recent study found that smartphones could be embodied into body schema by adapting the hand mental rotation task (Liu et al., 2022). In the present study, we adapted the hand mental rotation paradigm to explore the impact of smartphones on hand representation. We proposed that smartphone-related hand images would facilitate mental rotation.

Third, we adapted the forearm bisection paradigm (Garbarini et al., 2015; Romano et al., 2019; Sposito et al., 2012) to measure body metric representation and explore whether smartphone use extended the perceived forearm length. The forearm bisection task requires participants to estimate the perceived midpoint of their forearm (i.e., from the elbow to the tip of the middle finger) and thus access the perceived forearm length. Combined tool use training and the forearm bisection task, previous studies found that the perceived forearm length was longer after 20 min tool training than that before tool training, indicating that tool use extended forearm length representation (Sposito et al., 2012). In addition, brain-damaged hemiplegic patients with pathological deficits of embodiment also perceived a longer forearm length after 15 min tool training than before training (Garbarini et al., 2015). Besides, only imaging use of a tool instead of actual use could trigger tool incorporation into the body schema (Baccarini et al., 2014). Thus, we combined smartphone use imagery and the forearm bisection paradigm to explore the impact of smartphones on hand representation. We proposed that participants would perceive longer forearm length after smartphone use imagery than before.

The present study

According to the tool embodiment framework (Giummarra et al., 2008; Maravita & Iriki, 2004; Martel et al., 2016), tool use could update the encoding of the body schema, allowing the tool to be incorporated into the body representation and act as an extension of the body. Besides, the Extended-self theory (Belk, 1988, 2013) also suggested that the control we exert over external objects leads to self-extension, the perception of external objects as part of our body. Combining these two theories with previous research, we aimed to explore smartphone embodiment at two levels: The incorporation of smartphones into body schema and the extension of body representation. Specifically, we first adapted the SRC paradigm (Experiment 1) and the hand mental rotation paradigm (Experiment 2 and Experiment 3) to explore the unique role of smartphones in modifying the body schema compared to similarly shaped objects, traditional tools, and non-smartphones. Second, we adapted the forearm bisection paradigm (Experiment 4) to explore whether smartphone use could extend body metric representation.

Although a recent study provided an empirical basis for the smartphone embodiment by adapting the hand mental rotation task (Liu et al., 2022), our understanding of smartphone embodiment remains incomplete owing to four issues. In the present work, we addressed these issues and further explored the embodied effect of smartphones based on the tool embodiment framework.

First, comparing the mental rotation of smartphones and non-smartphones in Liu et al. (2022) cannot exclude the contribution of smartphone users' visual habituation of smartphone shapes to mental rotation. The integration of visual and motor-sensory information related to tool use plays an essential role in tool embodiment (Maravita & Iriki, 2004). Therefore, the influence of visual information (e.g., tool shape and size) needs to be considered when exploring the impact of smartphones on body representations from the tool embodiment perspective. In the present study, Experiment 1 and Experiment 2 aimed to explore whether the embodied effect of smartphones was independent of the smartphone-related shape and hand posture. In Experiment 1, we adapted the SRC paradigm and compared the SRC effects under two conditions (two types of pictures: a smartphone held in hand vs. a similarly shaped notebook held in hand). In both types of pictures, all factors (e.g., shape and size of the object and hand posture) are the same except for the handheld objects. Pictures with objects that are compatible with body schemas would trigger greater SRC effects or negative SRC effects. We hypothesized that:

H1a: The stimuli of pictures with a smartphone held in hand would trigger a larger SRC effect than the stimuli of pictures with a similarly shaped notebook held in hand (egocentric perspective in experiment 1a).

H1b: The stimuli of pictures with a smartphone held in hand would trigger a larger negative SRC effect (allocentric perspective in experiment 1b) than the stimuli of pictures with a notebook held in hand.

In Experiment 2, we adapted the hand mental rotation paradigm and compared the rotation performance under four conditions (four types of pictures: a smartphone held in hand, a similarly shaped notebook held in hand, a smartphone-use-related gesture without a smartphone, a smartphone-use-irrelated gesture). Participants were instructed to rotate these images and make a hand laterality judgment. Pictures with objects that are consistent with body schemas would facilitate the hand mental rotation. We hypothesized that:

H2: Compared to the stimuli of pictures with a similarly shaped notebook held in hand and a smartphone-use-irrelated gesture, the stimuli of pictures with a smartphone held in hand and smartphone-related gesture would facilitate the hand mental rotation.

Second, Liu et al. (2022) did not compare smartphones with traditional tools. Considering the previously mentioned differences between traditional tools and smartphones in terms of interaction with the human body, it remains unclear whether the embodied effect of smartphones is comparable to or greater than traditional tools. In the present study, Experiment 3 aimed to explore the unique role of smartphones in modifying the body schema compared to traditional tools and non-smartphones. In particular, we compared the rotation performance under four conditions (four types of pictures: A smartphone held in hand, a non-smartphone held in hand, a smartphone-use-related hand gesture without a smartphone, and a traditional tool held in hand). Considering the vital role of smartphones in modern human life, we hypothesized that:

H3: Compared to the stimuli of pictures with a non-smartphone held in hand and a traditional tool held in hand, the stimuli of pictures with a smartphone held in hand and a smartphone-related gesture would facilitate the hand mental rotation

Third, Liu et al. (2022) did not examine the extended effect of smartphone use on body representation. Besides being integrated into the body schema, could smartphones extend the body representation? In the present study, Experiment 4 aimed to further explore whether the embodied smartphones would extend forearm length representation. We adapted the forearm bisection task and compared the perceived forearm length before and after two imagination tasks (Imagine the context of using a smartphone and imagine the context of using a calculator). We hypothesized that:

H4: Compared to the perceived forearm length at baseline and after imagining using a calculator, the perceived forearm length would increase after imagining using a smartphone.

Finally, would the frequency of smartphone use affect the extent of smartphone embodiment? Given that tool embodiment is based on the experience of tool use (Giummarra et al., 2008; Jovanov et al., 2015; Osiurak & Federico, 2021; Weser & Proffitt, 2021), high-frequency smartphone users may accumulate more experience with smartphones and experience deeper smartphone embodiment. In the present study, we compared the SRC effect (Experiment 1), hand mental rotation performance (Experiment 2 and Experiment 3), and forearm length representation (Experiment 4) between high- and low-frequency smartphone users. Considering that tool-use experts have been found to be more tool-embodied than novices (Weser & Proffitt, 2021), we hypothesized that:

H5: The effects of smartphone embodiment would be larger among high-frequency smartphone users than that among low-frequency smartphone users in the above four experiments.

Experiment 1

Methods

Participants

Before the experiment, we distributed a questionnaire consisting of the Smartphone Addiction Scale for College Students (SAS-C) (Su et al., 2014) and a questionnaire of average daily smartphone use (Chen, 2016) to screen participants as high-frequency smartphone users and low-frequency smartphone users. SAS-C contains 22 items with six dimensions: Withdrawal behavior, salience behavior, social comfort, negative effects, use of the application, and renewal of App. All items were on a 5-point response scale (1 = *not at all true* to 5 = *always true*). Questionnaire of average daily smartphone use required participants to report their daily smartphone use time, the frequency of checking smartphones, and the smartphone use time before going to bed using a 5-point scale. A total of 458 questionnaire copies were randomly distributed and 450 questionnaire copies were collected. We invited participants who scored beyond the 27% distribution of either end of the tested population and built a participant pool. Except for individuals who did not provide contact information, the total number of participants in the participant pool was 212. All participants were recruited in this pool.

The target sample size of Experiment 1 was determined by considering the relevant observed effect sizes ($\eta_p^2 = 0.10$–0.70) in a previous study using the SRC paradigm (Vainio & Mustonen, 2011). Anticipating a medium to large effect size($\eta_p^2 = 0.10$), G*Power (Faul et al., 2009) estimated that a sample size of at least 10 participants per group would be necessary to detect an interaction of object type and smartphone use frequency with a power (1-β) of at least 80% at $\alpha = 0.05$. Thirty-nine undergraduates (mean age = 19.72, $SD = 1.34$) participated in Experiment 1a and forty-four undergraduates (mean age = 20.75, $SD = 1.73$) participated in Experiment 1b. Participants' grouping and SAS-C scores and the average daily smartphone use are presented in Table 1. All participants were healthy and had normal vision or corrected vision. All participants were given prior informed consent and received monetary compensation for their participation. The study was approved by the Ethics in Human Research Committee of the authors' University.

Materials and procedure

We photographed 22 pictures of a smartphone held in hand (smartphone-hand pictures). Before the experiment, we invited 44 college students to evaluate how the gestures in the pictures were consistent with their daily habits. Finally, the first and second most consistent gestures were chosen as stimuli. Based on the smartphone-hand pictures, we used Photoshop to replace the smartphone with a similarly shaped notebook while keeping the hand gestures to make the notebook-hand pictures (see Fig. 1). According to the stimuli features in previous studies (Vainio & Mustonen, 2011), the hands and the objects were in their natural color.

Table 1 Grouping of participants in four experiments

	Group	SAS-C Score			Average daily smartphone use		
		M	*SD*	*t*	*M*	*SD*	*t*
Experiment 1a	high-frequency ($n = 17$, 4 males)	86.71	4.62	16.63***	3.88	0.71	5.97***
	low-frequency ($n = 22$, 9 males)	46.59	9.06		2.62	0.61	
Experiment 1b	high-frequency ($n = 23$, 7 males)	89.43	5.88	20.76***	4.00	0.64	5.73***
	low-frequency ($n = 21$, 8 males)	49.30	6.94		2.75	0.80	
Experiment 2	high-frequency ($n = 20$, 5 males)	89.35	6.27	18.08***	3.97	0.63	4.83***
	low-frequency ($n = 19$, 6 males)	49.59	7.44		2.88	0.77	
Experiment 3	high-frequency ($n = 20$, 6 males)	87.50	6.42	15.09***	4.33	0.55	5.71***
	low-frequency ($n = 20$, 11 males)	51.15	8.65		3.27	0.63	
Experiment 4	high-frequency ($n = 20$, 4 males)	93.60	6.92	19.18***	4.07	0.61	4.76***
	low-frequency ($n = 20$, 6 males)	51.95	7.55		3.02	0.78	

SAS-C = Smartphone Addiction Scale for College Students; High-frequency = high-frequency smartphone users; low-frequency = low-frequency smartphone users. ***$p < 0.001$

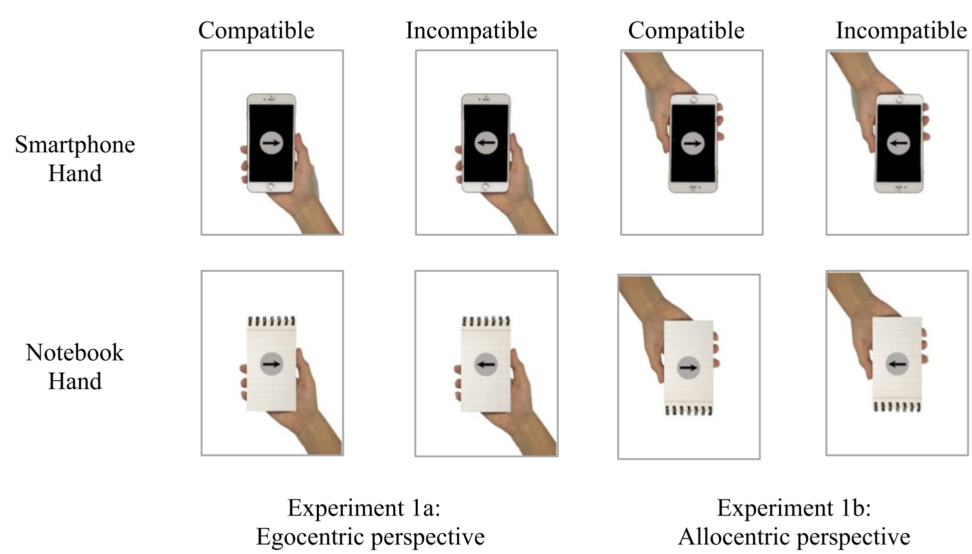


Fig. 1 Example stimuli in Experiment 1. Experiment 1a: Egocentric perspective, Experiment 1b: Allocentric perspective

The target stimuli were left–right pointing arrows. The black arrow was superimposed over a grey sphere to increase the arrow's discriminability from the background. The object-hand-arrow stimuli were displayed on a white background at the center of the screen. The stimuli were displayed from an egocentric viewpoint in Experinent1a, while the object-hand stimuli were displayed from an allocentric viewpoint in Experinent1b. The allocentric pictures were created by flipping vertically the object-hand images used in Experiment 1a. Example stimuli are presented in Fig. 1.

Each participant sat in a room with their head 60 cm in front of a computer display. An empty white screen was displayed for 1400 ms at the beginning of each trial. Then, the object-hand stimulus (a smartphone-hand picture or a notebook-hand picture) appeared on the screen. The onset of the target arrow varied among 100, 400, or 700 ms after the object-hand stimulus onset. The stimulus onset asynchrony (SOA) times were randomized between the trials. The arrow was displayed at the center of the screen, displaying for 200 ms. The participants were asked to press ";" on the keyboard after seeing the right-arrow and to press "a" on the keyboard after seeing the left-arrow as quickly as possible while maintaining accuracy. The object-hand stimulus remained on the screen until the participant had responded. If the participant did not respond within 1500 ms, the object-hand stimulus was removed from the screen. After the object-hand stimulus had disappeared from the screen, an empty white screen was displayed for 400 ms. Error responses were immediately followed by a red "×" displayed 200 ms on the screen. The experiment began with 20 practice trials followed by 480 formal trials, with a short break after 240 trials.

Data analysis

First, incorrect trials and correct trials with response times beyond two standard deviations from a participant's overall mean RT were excluded from RT analyses. Second, we calculated the SRC effect in smartphone-hand trials and notebook-hand trials. According to the previous study (Vainio & Mustonen, 2011), the difference value of RT or error rates between incompatible and compatible trials was defined as the SRC effect. Third, we conducted a 2 (group: high-frequency-smartphone users, low-frequency-smartphone users) × 2 (object type: smartphone, notebook) × 3 (SOA: 100 ms, 400 ms, 700 ms) repeated-measures ANOVA on SRC effect based on RT and error rates. All the statistical analyses were performed by SPSS 25.0. Greenhouse–Geisser correction of the ANOVA assumption of sphericity was applied where appropriate. The Bonferroni-corrected method was performed for post hoc testing of significant main effects, while simple effect analysis was used for testing significant interactions.

Results

Experiment 1a

In total, 5.1% of the raw data were discarded from the response time (RT) analysis, including 1.8% of incorrect trials and 3.3% of trials in which RTs were beyond two standard deviations from a participant's overall mean. The descriptive results of RTs and error rates in each condition are presented in Table S1 and S2, respectively. Table 2 presents the results of the repeated-measures ANOVA on the SRC effect. For the RT-based SRC effect, results revealed significant main effects of object type and SOA. The Bonferroni-corrected post-hoc pairwise comparisons revealed that the SRC effect in smartphone-hand trials ($M = 3.74$, $SE = 1.52$) was larger than that in notebook-hand trials ($M = -1.02$, $SE = 1.35$), $p < 0.01$. The SRC effect in the SOA condition of 400 ms ($M = 7.32$, $SE = 2.06$) was larger than that in the SOA condition of 100 ms ($M = -3.71$, $SE = 1.61$) and that in the SOA condition of 700 ms ($M = 0.48$, $SE = 1.96$), $ps < 0.01$. The difference between the SRC effect in the SOA condition of 100 ms and 700 ms was insignificant, $p = 0.36$. Additionally, results revealed a significant interaction of object type and SOA. Specifically, in the SOA condition of 400 ms, the simple effect of object type was significant, $F\ (1, 38) = 28.82$, $p < 0.001$, $\eta_p^2 = 0.431$; the SRC effect in smartphone-hand trials ($M = 12.62$, $SE = 2.51$) was larger than that in

Table 2 Repeated-measures ANOVA on SRC effect in Experiment 1a

Effect	Response time			Error rate		
	F(df)	p	η_p^2	$F(df)$	p	η_p^2
Object type	$F\ (1, 37) = 7.61$	0.009	0.17	$F\ (1, 37) = 0.18$	0.678	0.005
Group	$F\ (1, 37) = 0.52$	0.476	0.014	$F\ (1, 37) = 0.963$	0.333	0.025
SOA	$F\ (2, 74) = 9.29$	< 0.001	0.201	$F\ (2, 74) = 2.26$	0.112	0.057
Object type × Group	$F\ (1, 37) = 0.67$	0.418	0.018	$F\ (1, 37) = 0.03$	0.854	0.001
SOA × Group	$F\ (2, 74) = 0.05$	0.950	0.001	$F\ (2, 74) = 0.01$	0.987	< 0.001
Object type × SOA	$F\ (1.63, 60.41) = 4.27$	0.025	0.103	$F\ (2, 74) = 2.40$	0.098	0.061
Group × Object type × SOA	$F\ (1.63, 60.41) = 0.65$	0.495	0.017	$F\ (2, 74) = 0.13$	0.889	0.003

notebook-hand trials ($M = 1.69$, $SE = 2.46$). For error rates, results revealed no significant main effect and interaction.

Experiment 1b

In total, 5.3% of the raw data were discarded from the RT analysis, including 1.7% of incorrect trials and 3.6% of trials in which RTs were beyond two standard deviations from a participant's overall mean. The descriptive results of RT and error rates in each condition are presented in Table S3 and S4, respectively. Table 3 presents the results of the repeated-measures ANOVA on the negative SRC effect. For the RT-based negative SRC effect, results revealed significant main effects of object type and SOA. The Bonferroni-corrected post-hoc pairwise comparisons revealed that the negative SRC effect in smartphone-hand trials ($M = -6.46$, $SE = 1.32$) was larger than that in notebook-hand trials ($M = -1.89$, $SE = 1.41$), $p < 0.01$. The negative SRC effect in the SOA condition of 100 ms ($M = -8.38$, $SE = 1.86$) was larger than that in the SOA condition of 400 ms ($M = -1.44$, $SE = 1.51$), $p < 0.05$. The difference between the negative SRC effect in the SOA condition of 100 ms and that in the SOA condition of 700 ms was insignificant, $p = 0.11$. The difference between the negative SRC effect in the SOA condition of 400 ms and that in the SOA condition of 700 ms was insignificant, $p > 0.999$. For error rates, results revealed no significant main effect and interaction.

Discussion

Experiment 1 adapted the SRC paradigm to explore the effect of smartphones on body schema. The results supported H1a and H1b. We observed a larger SRC effect (egocentric perspective in experiment 1a) and a larger negative SRC effect (allocentric perspective in experiment 1b) in smartphone-hand trials compared to notebook-hand trials, indicating that participants showed a processing advantage for the picture of a hand holding a smartphone. These findings were consistent with the previous study that found the object SCR effect (Pellicano et al., 2010), which suggested that tool cues could initiate embodied simulations of tool use based on body schema, facilitating the processing of tools compatible with body schemas. Specifically, smartphone-hand pictures primed body schema and initiated embodied simulations, which facilitated the performance of compatible trials and interfered with the performance of incompatible trials. In contrast, the priming effect of notebook-hand pictures on body schema and embodied simulations was smaller than that of smartphone-hand pictures, thus leading to a smaller SCR effect in notebook-hand trials.

The processing advantage for hand images compatible with schema may be related to the function of the visuomotor system, which could automatically associate the visual objects with the hand motion schema to guide accurate movements (Shmuelof & Zohary, 2008a, b; Vainio & Ellis, 2020; Vainio & Mustonen, 2011). Moreover, object functional affordance was the critical factor in activating the body schema compatible with the object automatically (Buccino et al., 2009; Pellicano et al., 2010; Vainio & Ellis, 2020). Electrophysiological studies have shown that the connection of visual affordance and action occurs very early in the sensory pathways, reflecting the deep embodiment (Goslin et al., 2012). The strong functional affordance of smartphones may derive from the interaction between human hands and multifunctional smartphones, which makes smartphones deeply embodied in body schema. In contrast, the similarly shaped notebook held in hand lack of functional affordance and could not activate the body schema even with the same hand gesture. Overall, our findings extended previous studies and indicated that the smartphone-hand picture was more similar to the participants' body schema than the notebook-hand picture. Contrary to H5, the results showed no significant difference between the SRC effect of high-frequency smartphone use users and low-frequency smartphone use users regardless of the picture types.

Experiment 2

While the SRC paradigm involved the implicit priming of hand position perception, hand mental rotation required a conscious process involving integrating sensory information

Table 3 Repeated-measures ANOVA on negative SRC effect in Experiment 1b

Effect	Response time			Error rate		
	F(df)	p	η_p^2	$F(df)$	p	η_p^2
Object type	$F(1, 42) = 9.19$	0.004	0.180	$F(1, 42) = 3.01$	0.090	0.067
Group	$F(1, 42) = 2.15$	0.150	0.049	$F(1, 42) = 0.73$	0.397	0.017
SOA	$F(1.63, 68.56) = 5.37$	0.006	0.113	$F(2, 84) = 2.80$	0.067	0.062
Object type × Group	$F(1, 42) = 0.01$	0.910	< 0.001	$F(1, 42) = 1.40$	0.243	0.032
SOA × Group	$F(1.63, 68.56) = 0.31$	0.693	0.007	$F(2, 84) = 0.03$	0.967	0.001
Object type × SOA	$F(2, 84) = 0.25$	0.778	0.006	$F(2, 84) = 0.86$	0.426	0.020
Group × Object type × SOA	$F(2, 84) = 0.03$	0.971	0.001	$F(2, 84) = 1.69$	0.191	0.039

with motor movements. Hence, to further validate our findings in Experiment 1, we adapted the hand mental rotation paradigm in Experiment 2 to verify the impact of smartphones on hand representation.

Methods

Participants

The process of screening and recruiting participants was the same as Experiment 1. The target sample size was determined by considering the relevant observed effect sizes ($\eta_p^2 = 0.13$–0.76) in a previous study using the hand mental rotation task (Bonda et al., 1995; Ionta et al., 2007; Ter Horst et al., 2010). Anticipating a medium to large effect size($\eta_p^2 = 0.10$), G*Power (Faul et al., 2009) estimated that a sample size of at least 7 participants per group would be necessary to detect an interaction of picture type and user group with a power($1\text{-}\beta$) of at least 80% at $\alpha = 0.05$. Thirty-nine undergraduates (mean age = 20.38, *SD* = 1.65) from the participant pool built previously participated in Experiment 2. Participants' grouping was presented in Table 1.

Materials and procedure

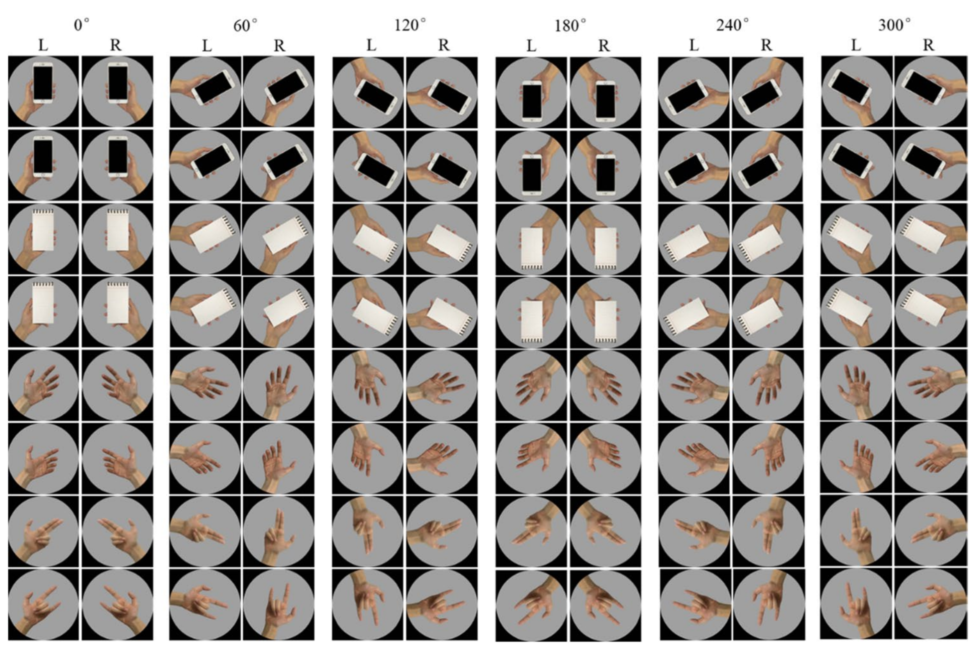
Four types of pictures of hand images were used as stimuli: (1) a smartphone held in hand (smartphone-hand); (2) a similarly shaped notebook held in hand (notebook-hand); (3) a smartphone-use related gesture without a smartphone (smartphone-related gesture); (4) a smartphone-use irrelated gesture (smartphone-irrelated gesture). The pictures used in Experiment 1a were adapted to make the four types of pictures. These four types of hand images were presented in 6 orientations defined by their clockwise rotation from the upright position of the smartphone: 0°, 60°, 120°, 180°, 240°, 300°. Example stimuli are presented in Fig. 2.

The experiment procedure was adapted from the previous studies (Bonda et al., 1995; Ionta et al., 2007; Ter Horst et al., 2010). Each participant was presented with a picture of the hand image in a particular orientation and decided whether the presented hand was the left or the right one. The interstimulus interval was 2 s. The participants were asked to press ";" on the keyboard after seeing the image of a right hand and to press "a" on the keyboard after seeing the image of a left hand as quickly as possible while maintaining accuracy. The participants were required to put their hands on the keyboard under the desk during the experiment. Each participant performed a set of 10 practice trials and four experimental blocks (96 trials each). Each hand image in both the left and right positions and in each orientation occurred randomly within each block.

Data analysis

First, incorrect trials and correct trials with response times more than 3600 ms and less than 300 ms were excluded from RT analyses according to the previous study (Ter Horst et al., 2010). Second, we conduct a 2 (user group: high-frequency smartphone users, low-frequency smartphone users) × 4 (picture type: smartphone-hand, notebook-hand, Smartphone-related gesture, Smartphone-irrelated gesture) × 6 (orientation: 0°, 60°, 120°, 180°, 240°, 300°) × 2 (hand laterality: left, right) repeated-measures ANOVA on correct mean RT and error rates. All the statistical analyses

Fig. 2 Stimuli of Experiment 2



were performed by SPSS 25.0. Greenhouse–Geisser correction of the ANOVA assumption of sphericity was applied where appropriate. The Bonferroni-corrected method was performed for post hoc testing of significant main effects, while simple effect analysis was used for testing significant interactions.

Results

In total, 3.9% of the raw data were discarded from the RT analysis, including 3.2% of trials that were errors and 0.7% of trials in which RTs were more than 3600 ms and less than 300 ms. The descriptive results of RT and error rates in each condition are presented in Table S5 and S6, respectively. Table 4 presents the results of the repeated-measures ANOVA on response time and error rates.

For RT, results revealed significant main effects of picture type and orientation. The Bonferroni-corrected post-hoc pairwise comparisons revealed that participants responded faster in smartphone-hand trials ($M = 835.52$, $SE = 40.77$) than in smartphone-irrelated gesture trials ($M = 867.54$, $SE = 41.78$), $p < 0.05$. Participants also responded faster in smartphone-related gesture trials ($M = 836.36$, $SE = 37.65$) than in smartphone-irrelated gesture trials, $p < 0.01$. There was no significant difference between RT in smartphone-hand trials and RT in notebook-hand trials ($M = 847.59$, $SE = 39.73$). There was no significant difference between RT in smartphone-hand trials and RT in smartphone-related gesture trials. The Bonferroni-corrected post-hoc pairwise comparisons revealed that the response time increased with the rotation angle from 0° to 180° (0°: 700.40 ± 29.75; 60°: 748.78 ± 33.16; 120°: 916.52 ± 48.17; 180°: 1080.54 ± 56.46, $ps < 0.001$). Then, response time decreased with the rotation angle from 180° to 300° (240°: 876.34 ± 44.52; 300°: 757.92 ± 34.38, $ps < 0.001$). There was no significant difference between RT in trials at relatively symmetrical angles (60° and 300°, 120° and 240°, $ps > 0.05$).

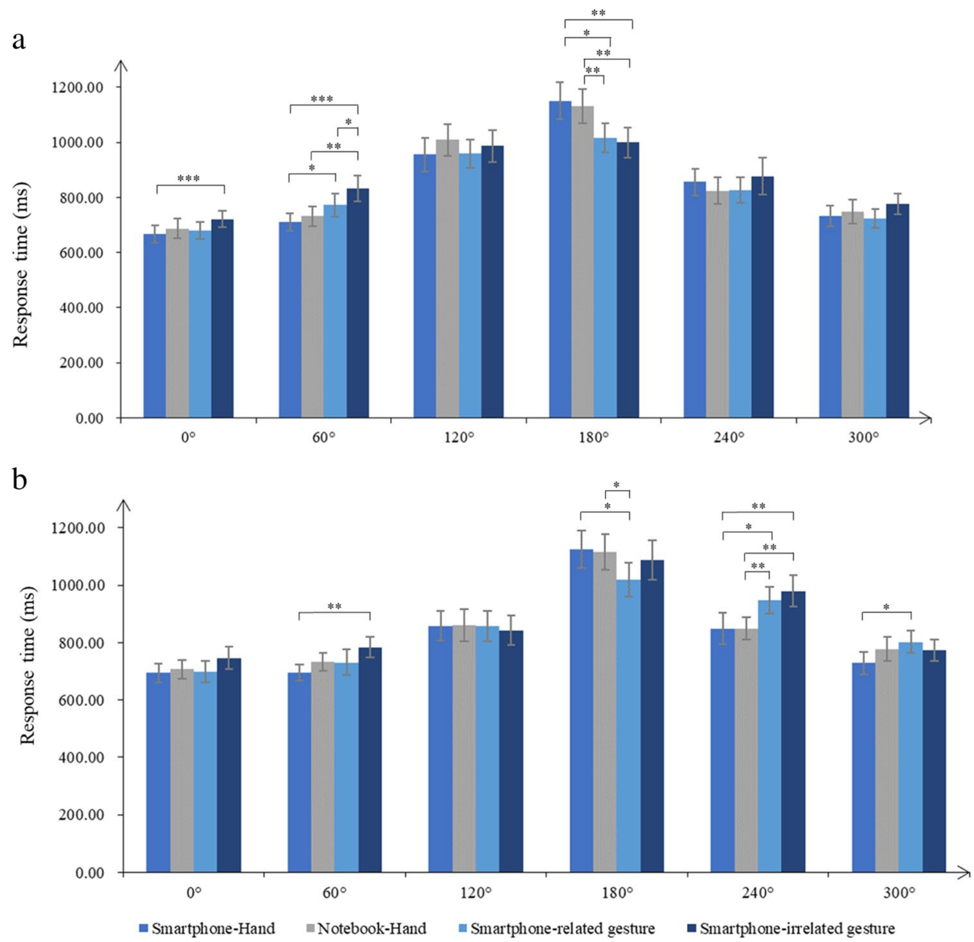
The results also revealed a significant three-way interaction among picture type, orientation, and hand laterality. Considering that images of left and right hands required different demands even though the rotate angle was the same, effects of picture type on RT were analyzed for images of left and right hands separately (see Fig. 3a for left hand stimuli and Fig. 3b for right hand stimuli). For left hand images, picture type showed significant simple effects in angle of 0°[$F(2.34, 88.88) = 3.21$, $p < 0.05$, $\eta_p^2 = 0.08$], 60° [$F(2.40, 91.23) = 13.03$, $p < 0.001$, $\eta_p^2 = 0.26$], and 180° [$F(3, 114) = 8.61$, $p < 0.001$, $\eta_p^2 = 0.19$]. For right hand images, picture type showed significant simple effects in angle of 60° [$F(2.14, 81.49) = 4.06$, $p < 0.05$, $\eta_p^2 = 0.10$], 180° [$F(3, 114) = 4.90$, $p < 0.001$, $\eta_p^2 = 0.11$], 240° [$F(2.41, 91.73) = 9.12$, $p < 0.001$, $\eta_p^2 = 0.19$], and 300° [$F(3, 114) = 3.22$, $p < 0.05$, $\eta_p^2 = 0.08$].

The Bonferroni-corrected Post-hoc test revealed that for left-hand images rotated 0°, the RT in the smartphone-hand trials ($M = 667.18$, $SE = 30.79$) was shorter than the RT in the smartphone-irrelated gesture trials ($M = 721.28$, $SE = 29.86$), $p < 0.001$; the differences among other types of images were insignificant. For left-hand images rotated 60°, the RT in the smartphone-hand trials ($M = 710.37$, $SE = 31.18$) was shorter than the RT in the smartphone-related gesture trials ($M = 772.74$, $SE = 42.70$), $p < 0.05$. The RT in the smartphone-hand trials was also shorter than

Table 4 Repeated-measures ANOVA on response time and error rates in Experiment 2

Effect	Response time			Error rate		
	F(df)	*p*	η_p^2	*F(df)*	*p*	η_p^2
Picture type	$F(2.47, 91.46) = 5.05$	0.005	0.120	$F(2.65, 98.12) = 0.33$	0.776	0.009
Group	$F(1, 37) = 0.10$	0.758	0.003	$F(1, 37) = 0.03$	0.858	0.001
Orientation	$F(1.68, 62.06) = 79.34$	< 0.001	0.682	$F(2.44, 90.15) = 28.42$	< 0.001	0.434
Hand laterality	$F(1, 37) = 0.10$	0.753	0.030	$F(1, 37) = 2.12$	0.154	0.054
Hand laterality × Group	$F(1, 37) = 0.05$	0.831	0.001	$F(1, 37) = 3.42$	0.073	0.084
Picture type × Group	$F(2.47, 91.46) = 0.08$	0.952	0.002	$F(2.65, 98.12) = 1.12$	0.34	0.029
Orientation × Group	$F(1.68, 62.06) = 1.15$	0.315	0.030	$F(2.44, 90.15) = 1.84$	0.157	0.047
Hand laterality × Picture type	$F(3, 111) = 1.28$	0.286	0.033	$F(3, 111) = 0.28$	0.842	0.007
Hand laterality × Orientation	$F(3.17, 117.17) = 7.81$	< 0.001	0.174	$F(3.34, 123.44) = 0.634$	0.611	0.017
Picture type × Orientation	$F(8.89, 328.86) = 6.52$	< 0.001	0.150	$F(7.44, 275.13) = 1.12$	0.349	0.029
Hand laterality × Picture type × Group	$F(3, 111) = 0.37$	0.773	0.010	$F(3, 111) = 0.99$	0.401	0.026
Hand laterality × Orientation × Group	$F(3.17, 117.17) = 1.22$	0.306	0.032	$F(3.34, 123.44) = 1.71$	0.162	0.044
Picture type × Orientation × Group	$F(8.89, 328.86) = 1.11$	0.356	0.029	$F(7.44, 275.13) = 1.81$	0.081	0.047
Hand laterality × Orientation × Picture type	$F(9.19, 339.91) = 2.20$	0.021	0.056	$F(7.59, 280.97) = 1.12$	0.348	0.029
Hand laterality × Orientation × Picture type × Group	$F(9.19, 339.91) = 1.28$	0.245	0.033	$F(7.59, 280.97) = 0.71$	0.671	0.019

Fig. 3 The interaction between picture type and rotation angle on response time for (**a**) left-hand stimuli and (**b**) right-hand stimuli



RT in the smartphone-irrelated gesture trials ($M = 832.74$, $SE = 47.03$), $p < 0.001$. The RT in the notebook-hand trials ($M = 731.57$, $SE = 34.53$) was shorter than the RT in the smartphone-irrelated gesture trials, $p < 0.01$. The RT in the smartphone-related gesture trials was also shorter than the RT in the smartphone-irrelated gesture trials, $p < 0.05$. For left-hand images rotated 180°, the RT in the smartphone-hand trials ($M = 1150.64$, $SE = 66.99$) was longer than the RT in the smartphone-related gesture trials ($M = 1015.89$, $SE = 52.84$), $p < 0.05$. The RT in the smartphone-hand trials was also longer than RT in the smartphone-irrelated gesture trials ($M = 1000.27$, $SE = 54.39$), $p < 0.01$. The RT in the notebook-hand trials ($M = 1130.35$, $SE = 62.22$) was longer than the RT in the smartphone-related gesture trials, $p < 0.01$. The RT in the notebook-hand trials was also longer than the RT in the smartphone-irrelated gesture trials, $p < 0.01$.

For right-hand images rotated 60°, the RT in the smartphone-hand trials ($M = 696.22$, $SE = 28.43$) was shorter than the RT in the smartphone-irrelated gesture trials ($M = 783.75$, $SE = 36.55$), $p < 0.01$; the differences among other types of images were insignificant. For right-hand images rotated 180°, the RT in the smartphone-hand trials ($M = 1124.86$, $SE = 64.41$) was longer than the RT in the smartphone-related gesture trials ($M = 1019.48$, $SE = 59.71$), $p < 0.05$. The RT in the notebook-hand trials ($M = 1114.84$, $SE = 61.93$) was also longer than the RT in the smartphone-related gesture trials, $p < 0.05$. For right-hand images rotated 240°, the RT in the smartphone-hand trials ($M = 848.61$, $SE = 54.07$) was shorter than the RT in the smartphone-related gesture trials ($M = 947.45$, $SE = 46.09$), $p < 0.05$. The RT in the smartphone-hand trials was also shorter than the RT in the smartphone-irrelated gesture trials ($M = 980.22$, $SE = 54.83$), $p < 0.01$. The RT in the notebook-hand trials ($M = 849.56$, $SE = 39.26$) was shorter than the RT in the smartphone-related gesture trials, $p < 0.01$. The RT in the notebook-hand trials was also shorter than the RT in the smartphone-irrelated gesture trials, $p < 0.01$. For right-hand images rotated 300°, the RT in the smartphone-hand trials ($M = 728.84$, $SE = 38.55$) was shorter than the RT in the smartphone-related gesture trials ($M = 802.68$, $SE = 39.04$), $p < 0.05$; the differences among other types of images were not significant.

For error rates, results revealed a significant main effect of orientation. The Bonferroni-corrected post-hoc pairwise comparisons revealed that the error rates increased with the rotation angle from 0° to 180° (0°: 0.45% ± 0.16%; 60°: 1.37% ± 0.36%; 120°: 5.19% ± 0.96%; 180°: 6.66% ± 0.98%, $ps < 0.05$). Then, the error rates decreased with the rotation angle from 180° to 300° (240°: 3.65% ± 0.54%; 300°: 1.25% ± 0.35%, $ps < 0.001$). There was no significant difference between error rates in trials at relatively symmetrical angles (60° and 300°, 120° and 240°, $ps > 0.05$).

Discussion

Experiment 2 adapted the hand mental rotation paradigm to verify the effect of smartphones on body schema. The results supported H2. We observed that participants' RT and error rates increased with rotation angle from 0° to 180° and decreased with the rotation angle from 180° to 300°, which was consistent with previous research on hand mental rotation (Ionta et al., 2007; Liu et al., 2022; Ter Horst et al., 2010). Moreover, compared to pictures of a smartphone-irrelated gesture, the smartphone-hand pictures facilitated the hand mental rotation for left-hand images rotated 0° and 60°, and right-hand images rotated 60°. In contrast, the smartphone-hand pictures interfered with the hand mental rotation for left-hand images rotated 180° and right-hand images rotated 240°. Compared to pictures of a smartphone-related gesture without a smartphone, the smartphone-hand pictures also facilitated the hand mental rotation for left-hand images rotated 60° and right-hand images rotated 300°. In contrast, the smartphone-hand pictures interfered with the hand mental rotation for left-hand images rotated 180° and right-hand images rotated 240°. Given that hand mental rotation involves simulating the hand and its movements (Ionta et al., 2007), hand mental rotation was accelerated when the viewed hand picture was similar in angle and posture to one's own hand body schema. Thus, our findings indicated that the smartphone-hand image was more similar to the participants' body schema than the hand gesture without a smartphone.

Contrary to the results of Experiment 1, the notebook-hand pictures showed a similar effect to the smartphone-hand pictures. However, the SCR paradigm adapted in Experiment 1 involved the implicit priming of hand position perception. The arrow target was superimposed on the hand-held objects, making the objects more prominent than the hand. In Experiment 2, the hand mental rotation paradigm required a conscious process involving integrating sensory information with motor movements (Ionta et al., 2007; Liu et al., 2022; Ter Horst et al., 2010), which made the hand position and gesture more prominent than the objects. Thus, the notebook-hand pictures showed a similar effect to the smartphone-hand pictures science the similar position and gesture. Contrary to H5, the results showed no significant difference between the hand mental rotation performance of high-frequency smartphone use users and low-frequency smartphone use users regardless of the picture types.

Experiment 3

In Experiment 1 and Experiment 2, we chose a similarly shaped notebook as the control stimulus for experimental control reasons, people actually do not often hold a notebook in the same position as a smartphone, and a notebook is not a commonly used handheld tool. Thus, in Experiment 3, we added pictures of a traditional tool and a non-smartphone held in hand to further validate the embodiment of smartphones.

Methods

Participants

The process of screening and recruiting participants was the same as Experiment 2. Forty undergraduates (mean age = 17.92, $SD = 0.27$) participated in Experiment 3. Participants' grouping was presented in Table 1.

Materials and procedure

In addition to the picture types used in Experiment 2, we photographed 35 pictures of hand images consisting of a smartphone (i.e., iPhone, Android phone) held in hands, a non-smartphone (i.e., Nokia slide phone, TCL flip phone) held in hand, smartphone-use related gestures, and a traditional tool (i.e., spoon, pen, chopsticks) held in hands. We chose spoons, pens, and chopsticks as alternative traditional tools given they are similar in size to smartphones and are used frequently in students' daily lives. In addition, the sense of reward and control is the key to incorporating objects into the self (Christoff et al., 2011; Zhou et al., 2015). Spoons, pens, and chopsticks are used to obtain something, which may bring some reward and a sense of control. Therefore, we selected these three alternative traditional tools. We then invited 82 participants to evaluate the use frequency and familiarity of the handheld tools on the alternative pictures and finally chose four types of pictures that conformed to their habits: (1) a smartphone held in hand (smartphone-hand); (2) a non-smartphone held in hand (non-smartphone-hand); (3) a smartphone-use related gesture without a smartphone (smartphone-related gesture); (4) a spoon held in hand (spoon-hand). These four types of hand images were presented in four orientations defined by their clockwise rotation from the upright position of the smartphone: 0°, 90°, 180°, 270°. Example stimuli are presented in Fig. 4.

The procedure of Experiment 3 was similar to that of Experiment 2. Each trial started with a fixation point

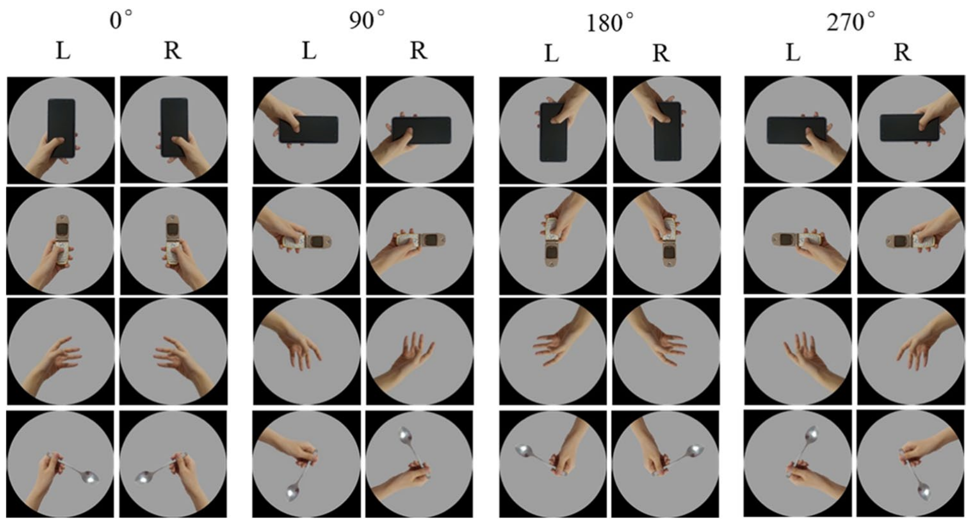


Fig. 4 Stimuli of Experiment 3

presenting 1000–1500 ms randomly, followed by a randomly selected stimulus of a hand image presented until the participant responded. The participants were asked to press “;” on the keyboard after seeing the image of a right hand and to press “a” on the keyboard after seeing the image of a left hand as quickly as possible while maintaining accuracy. Based on previous studies (Liu et al., 2022; Lyu et al., 2016; Yu et al., 2020) and the results of Experiment 2, no significant differences were found for symmetric orientations (i.e., 60 vs. 300 and 90° vs. 270°). Thus, we present the images with orientation angles of 0°, 90°, 270°, and 180° in the ratio of 2:1:1:2, and collapsed the data of symmetric orientations (i.e., 60 vs. 300) in data analysis. Each participant performed a set of 10 practice trials and ten experimental blocks (96 trials each) with several self-determined inter-block breaks.

Data analysis

The data analysis process in Experiment 3 was similar to that in Experiment 2.

Results

In total, 3.9% of the raw data were discarded from the RT analysis, including 1.6% of errors and 0.8% of trials in which RTs were more than 3600 ms and less than 300 ms. The descriptive results of RT and error rates in each condition are presented in Table S7 and S8, respectively. Table 5 presents the results of 2 (user group: High-frequency smartphone users, low-frequency smartphone users) × 4 (picture type: Smartphone-hand, non-smartphone-hand,

Table 5 Repeated-measures ANOVA on response time and error rates in Experiment 3

Effect	Response time			Error rate		
	F(df)	p	η_p^2	$F(df)$	p	η_p^2
Picture type	F (1.76, 67.03) = 4.33	0.02	0.102	F (2.55, 97.04) = 1.40	0.250	0.036
Group	F (1, 38) = 0.053	0.819	0.001	F (1, 38) = 0.15	0.705	0.004
Orientation	F (1.36, 51.68) = 125.89	< 0.001	0.768	F (1.26, 47.85) = 23.22	< 0.001	0.379
Hand laterality	F (1, 38) = 10.77	0.002	0.221	F (1, 38) = 1.95	0.172	0.049
Hand laterality × Group	F (1, 38) = 1.07	0.307	0.027	F (1, 38) = 0.76	0.390	0.020
Picture type × Group	F (1.76, 67.36) = 0.76	0.457	0.020	F (2.55. 97.04) = 0.69	0.540	0.018
Orientation × Group	F (1.36, 51.68) = 0.09	0.835	0.002	F (1.26, 47.85) = 0.480	0.535	0.012
Hand laterality × Picture type	F (3, 114) = 1.78	0.156	0.045	F (3, 114) = 1.01	0.390	0.026
Hand laterality × Orientation	F (1.55, 58.76) = 3.38	0.053	0.082	F (4.34, 53.70) = 0.31	0.659	0.008
Picture type × Orientation	F (2.72, 103.23) = 0.85	0.463	0.022	F (3.97, 150.94) = 1.72	0.149	0.043
Hand laterality × Picture type × Group	F (3, 114) = 0.25	0.862	0.006	F (3, 114) = 1.65	0.183	0.042
Hand laterality × Orientation × Group	F (1.55, 58.76) = 0.25	0.725	0.006	F (1.41, 53.70) = 0.27	0.687	0.007
Picture type × Orientation × Group	F (2.72, 103.46) = 0.99	0.394	0.025	F (3.97, 150.94) = 0.70	0.594	0.018
Hand laterality × Orientation × Picture type	F (4.18, 158.89) = 0.55	0.710	0.014	F (3.10, 117.72) = 1.77	0.154	0.045
Hand laterality × Orientation × Picture type × Group	F (4.18, 158.89) = 0.99	0.417	0.025	F (3.10, 117.72) = 0.70	0.556	0.018

smartphone-related gesture, spoon-hand) × 3 (orientation: 0°, 90°, 180°) × 2 (hand laterality: Left, right) repeated-measures ANOVA on RT and error rates.

For RT, results revealed significant main effects of picture type, hand laterality, and orientation. The Bonferroni-corrected post-hoc pairwise comparisons revealed that participants responded faster in smartphone-hand trials ($M = 838.88$, $SE = 36.82$) than in non-smartphone-hand trials ($M = 869.64$, $SE = 37.42$, $p < 0.001$) and spoon-hand trials ($M = 876.29$, $SE = 39.19$, $p < 0.01$). There was no significant difference between RT in Smartphone-hand trials and RT in Smartphone-related gesture ($M = 860.29$, $SE = 39.19$), $p = 0.39$. Participants also response faster for right hands ($M = 839.37$, $SE = 38.44$) than for left hands ($M = 839.37$, $SE = 38.44$). The Bonferroni-corrected post-hoc pairwise comparisons revealed that the response time increased with the rotation angle from 0° to 180° (0°: 724.70 ± 33.95; 90°: 835.22 ± 40.56; 180°: 1024.10 ± 45.69, $ps < 0.001$).

For error rates, results revealed a significant main effect of orientation. The Bonferroni-corrected post-hoc pairwise comparisons revealed that the error rates increased with the rotation angle from 0° to 180° (0°: 0.32% ± 0.08%; 90°: 1.15% ± 0.26%; 180°: 3.44% ± 0.63%, $ps < 0.05$).

Discussion

As a complement to Experiment 1 and Experiment 2, Experiment 3 compared the effects of a traditional tool, a non-smartphone, and a smartphone on the hand mental rotation to validate the embodiment of smartphones further. The results supported H3 and indicated that compared to spoon-hand pictures and non-smartphone-hand pictures, the pictures of a smartphone held in hand facilitated the hand mental rotation. These findings contributed to the evidence that smartphones had a greater impact on the body schema than traditional tools (e.g., spoons) as well as non-smartphones, highlighting the unique embodiment effect of smartphones. In line with the results of Experiment 1 and Experiment 2, the results of Experiment 3 did not show any significant difference between high-frequency smartphone users and low-frequency smartphone users.

Experiment 4

Experiment 4 combined smartphone use imagery and the forearm bisection paradigm to further explore whether the embodied smartphone would lead to the extension of perceived forearm length representation.

Methods

Participants

The process of screening and recruiting participants was the same as Experiment 1. The target sample size of Experiment 4 was determined by considering the relevant observed effect sizes ($\eta_p^2 = 0.07–0.92$) in a previous study (Sposito et al., 2012) using the forearm bisection paradigm. Anticipating a medium to large effect size($\eta_p^2 = 0.10$), G*Power (Faul et al., 2009) estimated that a sample size of at least 9 participants per group would be necessary to detect an interaction among the factors with a power(1-β) of at least 80% at $\alpha = 0.05$. Forty undergraduates (mean age = 20.00, $SD = 1.72$) participated in Experiment 4.

Materials and procedure

We manipulated the perception of smartphone function by developing two imagination tasks. One was the imagination of using smartphones (experimental task), which was aimed to prime the feeling of using smartphones. The control task was the imagination of using a calculator, which was similar in gesture and movement to using a smartphone. In the smartphone-use imagination task, participants were required to hold the smartphone and imagine chatting with a friend and browsing social networking sites. In the calculator-use imagination task, participants were required to hold the calculator and calculate the equation (only addition and subtraction) that the experimenter was reporting. Throughout the two imagination tasks, the participants were blindfolded and were asked to focus on the contents in the imagination and the movement of their hands and fingers.

To test the effectiveness of manipulation, we developed a questionnaire to assess the feeling during the imagination tasks. Based on the feature of a smartphone (i.e., ease of social connection, network accessibility, and the aggregation of various functions), we developed six items (0–10 Likert scale) to measure the perceived connection with others and the outside world. The six items were: “To what extent do you think you are connected to others?”, “How connected do you think you are to others?”, “How far do you think you are from others?”, “To what extent do you think you are connected to the world?”, “How much of the world do you think you can catch?”, and “How far do you think you are from the world?”. The third and the sixth item had to be scored in reverse. The participants filled out the questionnaire three times (baseline and after each imagination task). Cronbach’s alphas for the questionnaire three times were 0.82, 0.86, and 0.86.

The bisection task was adapted from previous studies (Liu et al., 2019; Sposito et al., 2012). Blindfolded participants sat on a chair with their right forearm placed radially on a plain surface located 30 cm rightwards from the midsagittal plane. On each trial, participants were instructed to point out the midpoint of their right upper limb segment comprising the forearm and the hand, considering the elbow and the tip of the middle finger as the two extremities. A transparent screen was placed about 3 cm above the arm during the bisection task to avoid tactile feedback. There was no time constraint, but corrections were not allowed. On each bisection session, a paper rule was placed on top of the plastic screen, with the 0-cm point in correspondence of the tip of the middle finger, to measure the position of the subjective midpoint (p). The point p was measured to the nearest millimeter starting from the 0-cm point. Afterward, a percentage score relative to the participant's forearm length was calculated using the formula: [(p/arm length) × 100%]. A value higher than 50% indicated a deviation of the subjective midpoint towards the elbow (i.e., proximal deviation). In comparison, a value lower than 50% indicated a deviation towards the hand (i.e., distal deviation). Participants performed 45 trials (15 for baseline and 15 after each of the two imagination tasks).

Participants performed tasks in two stages. Stage 1 was the baseline stage; in this stage, participants were required to fill out the manipulation check questionnaire and then perform the bisection task. Stage 2 was the task stage; participants were required to perform the smartphone-use imagination task or the calculator-use task. After each imagination task, participants were required to fill out the manipulation check questionnaire and then perform the bisection task again. The order of the two tasks was counterbalanced between participants.

Data analysis

We conduct a 2 (participant type: high-frequency-smartphone users, low-frequency-smartphone users) × 3 (imagination task: baseline, smartphone use, calculator use) repeated-measures ANOVA on the scores of the manipulation check questionnaire and the subjective midpoints of forearms. All the statistical analyses were performed by SPSS 25.0. Greenhouse–Geisser correction of the ANOVA assumption of sphericity was applied where appropriate. The Bonferroni-corrected method was performed for post hoc testing of significant main effects, while simple effect analysis was used for testing significant interactions.

Results

The repeated-measures ANOVA on the scores of the manipulation check questionnaire revealed a significant main effect of task type, $F(1.52, 57.87) = 18.37$, $p < 0.001$, $\eta_p^2 = 0.33$. The Bonferroni-corrected post-hoc pairwise comparisons revealed that the score of the manipulation check questionnaire after the smartphone-use imagination task ($M = 6.15$, $SE = 0.24$) was higher than the score on the baseline ($M = 5.07$, $SE = 0.28$), $p < 0.01$. The score after the smartphone-use imagination task was also higher than the score after the calculator-use imagination task ($M = 4.04$, $SE = 0.33$), $p < 0.01$. In contrast, the score of the manipulation check questionnaire after the calculator-use imagination task was lower than the score on the baseline, $p < 0.01$.

The repeated-measures ANOVA on the subjective midpoints of forearms revealed a significant main effect of task type, $F(2, 76) = 5.60$, $p < 0.01$, $\eta_p^2 = 0.13$. The Bonferroni-corrected post-hoc pairwise comparisons revealed that the proximal deviation after the smartphone-use imagination task ($M = 52.61\%$, $SE = 1.24\%$) was smaller than the baseline proximal deviation ($M = 55.16\%$, $SE = 1.02\%$), $p < 0.01$. The proximal deviation after the smartphone-use imagination task was also smaller than that after the calculator-use imagination task ($M = 54.44\%$, $SE = 1.18\%$), $p < 0.01$. There was no significant difference between the proximal deviation at baseline and that after the calculator-use imagination task, $p > 0.05$.

Discussion

Based on the results of Experiment 1 to Experiment 3 showing the advantages of processing smartphone-use-related body schema, Experiment 4 further explored the extended effect of smartphone use on perceived forearm length representation. The results supported H4. We found that the participants' baseline level of perception of the connection with others and the outside world was moderate. After imaging the use of smartphones, the level increased, while the level decreased after imaging the use of calculators. Moreover, we also found that participants' perceived forearm length increased after imaging smartphone use compared to that at baseline and that after imaging using a calculator. These findings indicated that the smartphone-use imagination had primed the feeling of connecting oneself to other people and the outside world and increased the representation of forearms length.

These findings provide direct empirical evidence for the extended self theory (Belk, 1988, 2013), which suggests the perceived sense of extended self while exercising power or control over external objects. Besides, the self also extends into the digital world (e.g., virtual avatars and narratives of self on social media networks) as the environment and behavior of modern human life change (Belk, 2013). In our study, the sense of extended self may be triggered during the smartphone-use imagination task and be embodied as the extended forearm length representation.

These findings are in line with the tool embodiment effect found in traditional tools, which suggests that tool use could induce changes in body plasticity, leading to the integration of tools into body representations and the extension of body representations (Baccarini et al., 2014; Garbarini et al., 2015; Romano et al., 2019; Sposito et al., 2012). Moreover, the present findings expanded the effect to modern tools (i.e., smartphones), indicating smartphone embodiment. Although the length of a smartphone was much shorter than that of a traditional tool (e.g., a rake) and the smartphone did not allow for a physical grasping motion, we found that the sense of access and connection during smartphone use could be embodied in an extension of the body length representation. Contrary to H5, the forearm length perceived by high-frequency smartphone users was not longer than that perceived by low-frequency smartphone users.

General discussion

In the current study, we explored the association between smartphone use and body representation from the perspective of tool embodiment. By adopting the paradigms of hand representation (Bonda et al., 1995; Ionta et al., 2007; Vainio & Mustonen, 2011) and body extension after tool use (Liu et al., 2019; Sposito et al., 2012), we found that smartphones could be integrated into body schema and lead to the extension of perceived forearm length.

Smartphones could be integrated into the body schema

Experiment 1 to Experiment 3 provided evidence that participants showed a processing advantage for the picture of a smartphone held in hand, indicating that smartphones could be integrated into individuals' body schema. This integration might be the result of an overlearning of the hand-smartphone association. Hand representation is extensive and overlearned; thus, the identity is automatically and rapidly extracted from a static image of a hand and mapped to the corresponding hand representation (Vainio & Mustonen, 2011). The visuomotor mechanisms that extract the hand identity and bind it with one's corresponding motor representations appear to operate in a remarkably over-learned manner. These processes operate rapidly and implicitly and can be triggered by a very brief viewing of an image of a static hand shaped in different postures (Vainio & Mustonen, 2011). Thus, the representation of smartphone-hand pictures may be similar to such an extensive and overlearned process of hand representation. Although smartphones are external objects, using experience may enhance the visuomotor mechanisms of hand-related representation in a remarkably over-learned manner.

Additionally, the proprioceptive information accumulated while using smartphones may also contribute to smartphone embodiment. Accurate and timely representation of body parts in space requires integrating visual and proprioceptive information (Assaiante et al., 2014; Dijkerman & de Haan, 2007). The visual information relies on visual cues from the viewed hand and the proprioceptive information involves the body experience of the participants. Previous studies comparing the mental rotation of hands and feet have found that the hands accumulated richer body experience than the feet and formed more proprioceptive information, facilitating the mental rotation process (Ionta et al., 2007). Thus, the accumulated smartphone-related proprioceptive information during hand-smartphone interaction is the key to generating the smartphone-related hand representation.

Smartphone use could extend body representation

Experiment 4 provided evidence that smartphone use led to the extension of the forearm length representation. These findings suggested that smartphone use showed a similar effect on body representation as a traditional tool (Cardinali et al., 2009; Sposito et al., 2010, 2012), even though smartphones had no obvious extension of physical length. Previous traditional tool-use studies involved the training of using a long tool such as a 60 cm rake to capture some objects far away and then extended the representation of hands length (Sposito et al., 2012). Why did smartphones show a similar effect even though they lacked apparent physical length? It could be explained from three perspectives.

First, tool-use after-effects generalized to pointing movements, despite the absence of specific tool training (Cardinali et al., 2009). That is, only functional sense (e.g., gaining) without accurate and successful control of body movements (e.g., real grasp) was enough to alter body representation. Indeed, only imaging use of a tool without actual use could trigger tool incorporation into the body schema (Baccarini et al., 2014). Therefore, the imagination of using smartphones primed the gaining of social connection and information, which was enough to extend the body representation. Second, holding a smartphone reminded the connection to the world with just one click. These feelings were related to the sense of agency, which can help us identify our body self (Christoff et al., 2011; Franklin & Wolpert, 2011) and facilitate the incorporation of external objects into the self. Third, interpersonal distance and information accessibility primed by smartphone use were abstract and could be understood by conceptual metaphor, mapping a concrete concept based on body experience (Landau et al., 2010; Yin et al., 2013). Therefore, body representation extension via smartphone use may result from the embodied understanding of the primed abstract feelings.

The role of smartphone use frequency on smartphone embodiment

We also hypothesized more prominent smartphone embodiment among high-frequency smartphone users than low-frequency smartphone users. However, the results of the four experiments showed no significant difference between high-frequency smartphone users and low-frequency smartphone users, indicating that the effect of smartphone use on body representation might reach a ceiling. A possible explanation is that the popularity and indispensability of smartphones do not allow low-frequency use, which leads to the ceiling of smartphone use. Instant tool use training (i.e., 20 min) can temporarily modify body representation (Sposito et al., 2012). Moreover, a pointing movement can lead to similar effects of tool use training (Cardinali et al., 2009). Therefore, daily smartphone use experience could be enough to impact body representation.

Another possible explanation is that high-frequency smartphone users may have a higher smartphone addiction tendency, which is found related to a general deficit in inhibitory control (Chen et al., 2016). And addictive clues were distractions for attention during cognitive tasks (Jeromin et al., 2016a, b). Therefore, the processing advantage of smartphone-related hand images may be counteracted by the deficit in the cognition process among high-frequency smartphone users, thus leading to the same response pattern as low-frequency smartphone users.

Implications and limitations

Our study makes three contributions by expanding upon and complementing previous findings on tool embodiment. First, the current findings provided empirical evidence for extended-self theory (Belk, 1988, 2013) applied to smartphones, which has been investigated theoretically and qualitatively (Clayton et al., 2015; Hoffner et al., 2016; Park & Kaye, 2019). Second, the present findings expand the research fields of tool embodiment from traditional tools (Jovanov et al., 2015; Martel et al., 2016, 2021; Sposito et al., 2012) to the modern technological tool (i.e., smartphones). We found that smartphones had a similar effect as traditional tools on body representation, broadening the conceptual view of tool embodiment. Third, we verified the previous findings of smartphone embodiment (Liu et al., 2022) by comparing the effects of smartphones on body representation with that of a similarly shaped notebook and a traditional tool with similar use frequency. Indeed, we underscored the unique role of smartphones in modifying the body schema, which was more potent than both traditional tools and non-smartphones. From a practical standpoint, our study also provides human factors related to embodied cognition for designing smartphones or other wearable devices. Enhancing the multisensory interaction of smart tools with humans and thus promoting smart tools embodied in the human body may improve the ease of use, enjoyment, and continued use of smart tools (Shaw et al., 2018). Besides, smartphone embodiment might underlie the impact of smartphone use on humans, which may provide a new perspective on prevention and interventions for smartphone addiction.

The current study has certain limitations that should be noted. First, given that our participants were college students, their smartphone experience might be rich regardless of the frequency of using smartphones. It is also one possible reason for the ceiling of smartphone use. Future research should consider sampling and recruiting participants with few smartphone experiences, such as adolescents from rural areas. Second, it may be challenging to find a handheld tool that rivals the smartphone in terms of familiarity, frequency of use, and usage posture to serve as a control stimulus. The control materials are limited to the notebook, spoon, and calculator in the present study. Future research could consider more handheld tools and control their familiarity to clarify the special effects of smartphones. Third, the present study only provided primary evidence for the association between smartphone use and body representation. More replicate studies and mechanism explorations are needed. The sense of ownership and agency is essential for body recognition and representation (Christoff et al., 2011; D'Angelo et al., 2018), as well as for human–computer interaction (Shaw et al., 2018). Thus, future research could consider the role that the sense of agency and ownership played in the association between smartphone use and body representation.

Conclusion

The present study explored the association between smartphone use and body representation. We found that the pictures of a smartphone held in hand facilitated hand recognition and representation, which meant that smartphones were integrated into body representation. Moreover, we also found that the perceived forearm length of the participants increased after imagining using a smartphone, indicating that smartphone use led to the extension of the forearm length representation. However, we found no significant difference between high-frequency smartphone users and low-frequency smartphone users, indicating that the effect of smartphone use on body representation reached a ceiling. These findings broadened the conceptual view of tool embodiment to smartphone embodiment and provided a novel insight into the impact of smartphone use on humans from the perspective of tool embodiment.

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Data and materials availability All data and materials used in this study are included in this published article and its supplementary information files.

Declarations

Ethics approval The study was approved by the Ethics in Human Research Committee of Central China Normal University.

Consent All participants were given prior informed consent and received monetary compensation for their participation.

Competing interests The authors declare that they have no conflict of interest.

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